

Weekly Lesson Plan – Week 6 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon Week of: September 14, 2026 Course: Engineering Design & Problem Solving (InSPIRESS)				
TEKS / ELPS:				
TEKS: §127.406(4) communicate findings and proposed solutions using appropriate tools; §127.406(6) implement a design process to develop a multi-system product or solution ELPS: c.1C, c.2C, c.3E, c.4C, c.4F, c.5G				
Aim of the Lesson / Objective: <i>Statement of what you want students to do (skill) and/or know (knowledge). Students will...</i>				
MONDAY SEP 14	TUESDAY SEP 15	WEDNESDAY SEP 16	THURSDAY SEP 17	FRIDAY SEP 18
Students will create their Autodesk education account with their school email, join the class hub, sign in to Fusion 360, and locate their private work folder.	Students will sketch a dimensioned rectangle, extrude it into a solid, and use a rectangular pattern to create a toy block with eight studs.	Students will apply shell, extrude, and fillet features to complete the toy block model and save it to their work folder.	Students will insert and calibrate a reference image, sketch a profile with lines and a spline, revolve it into a 3D bottle, and finish it with shell, fillet, and a glass appearance.	Students will complete the toy block model and save it to their work folder, hand the computer off to a classmate who has not started, and complete a CAD vocabulary word search while waiting.
Employed Language Domain(s) Objective: <i>How students achieve the objective – Listening, Speaking, Reading, Writing. Students will...</i>				
Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members' ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings.				
Assessment of Content Objective: <i>Instrument used to determine mastery. Ex: Exit Ticket.</i>				
Students will be assessed by written, group, or oral responses throughout the class period.				
Set (Bell Work): <i>A motivating action that engages students in the objective – question, fact, activity. Essential Question:</i>				
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Your company can make its product \$2 cheaper by switching to a plastic part that wears out in about 3 years instead of 10. Customers will not notice at first. Do you make the switch?	A LEGO brick from 1958 still snaps onto one made today. What has to be true about the measurements for that to work?	Most plastic parts are hollow on the inside, not solid. Why would a company make a part hollow?	Name something in your house that is round and could be made by spinning one flat outline around a center line. Describe the outline.	Your team finds a small flaw one week before your product launches. Fixing it delays the launch a month and costs the team its bonus. The flaw might hurt 1 person in 100,000. What do you do?
Methodology / Steps: <i>What it takes to teach the objective so 100% can pass the assessment. Teacher will... Teams will... Individuals will...</i>				
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Step 1: Bell ringer / essential question. Step 2: Bell work (dilemma): cheaper part that wears out sooner - make the switch? (10 min) Step 3: How today works: five steps, in order (3 min) Step 4: Steps 1-2: school Outlook, then Join Hub (makes your one account) (10 min) Step 5: Step 3: open Fusion (desktop or browser) and sign in (7 min) Step 6: Steps 4-5: switch to the class hub, find your folder (10 min) Step 7: Check-out: teacher sees your folder; finishers help a neighbor (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell work: why do LEGO measurements have to be exact? (10 min) Step 3: New file saved in your folder; how we watch: pause, build, play (5 min) Step 4: Day 1 video: sketch, extrude, stud, rectangular pattern (25 min) Step 5: Save (Ctrl+S) before the bell (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell work: why are most plastic parts hollow? (10 min) Step 3: Finish Day 1: shell through the bottom, inside columns, fillet (27 min) Step 4: Save over yesterday's file (3 min) Step 5: Finished? Start Day 2 early (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell work: what in your house is made by spinning a shape? (10 min) Step 3: Save the bottle picture; new file saved in your folder (5 min) Step 4: Day 2 video: canvas, calibrate, spline, revolve, shell, fillet, glass (28 min) Step 5: Save (Ctrl+S) before the bell (2 min)	Step 1: Bell ringer / essential question. Step 2: Bell work (dilemma): a flaw found a week before launch - delay or ship? (10 min) Step 3: Finish the Toy Block (shell, inside columns, fillet); students waiting start the word search (25 min) Step 4: Hand-off: save, sign out, pass the computer to the next student (8 min) Step 5: Check: file saved in Students > your name; word search turned in (2 min)
Resources: <i>Textbook, handouts, visuals, laptops.</i>				
Google Classroom, projector, engineering notebook, Design Brief packet / worksheets, laptops or phones for AI tools.				
Re-Teaching Activities: <i>What you prepare if students do not master the objective.</i>				
Review during bellwork; small-group re-teach at the teacher table; one-on-one explanation on request; peer and native-language support within the seating group.				
Reflection: <i>Students analyze what they accomplished and how they reached mastery.</i>				
Exit ticket – students state what they accomplished and what worked best.				
Study Skills: <i>Skills implemented by students – note-taking, summarizing, teamwork, etc.</i>				
Note-taking, summarizing, outlining, reading comprehension, teamwork, oral communication, problem-solving.				
SPED & 504 Accommodations / Modifications: <i>Codes.</i>				
Per Individualized Education Plan (IEP) / 504 plan.				
ELL Accommodations: <i>Codes.</i>				
Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: students are seated in mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context, rehearse with a peer before whole-class response, and can be pulled as a small group for re-teach (PN, WT). ONE-ON-ONE: individual teacher explanation is available on request every period; directions are restated and modeled individually at the student desk after whole-class delivery (CD, RS, VC).				